

INSTRUMENT

FLIGHT PROCEDURES

INSTRUMENT FLIGHT PROCEDURES

- **1. Introduction to IFP**
- **2. Arcs & Radials**
- **3. Fix to fix Nav**
- **4. Holding**
- **5. Instrument Approach Plates**
- **6. Instrument Approaches**
- **7. Over-run/Revision**
- **8. Exam**

INTRODUCTION

THE AIM OF IF IS TO FLY SAFELY USING THE A/C INSTRUMENTS UNDER IMC CONDITIONS.

IMC: Instrument Meteorological Conditions.

VMC: Visual Meteorological Conditions.

IFR: Instrument Flight Rules.

VFR: Visual Flight Rules.

AIRSPACE CLASSIFICATION

- **CLASS A:**
IFR FLIGHTS ARE ONLY PERMITTED, AND SUBJECT TO AIR TRAFFIC CONTROL SERVICE AND ARE SEPARATED FROM EACH OTHER.
- **CLASS B: (NOT USED IN THE MUSCAT FIR)**
IFR AND VFR FLIGHTS ARE PERMITTED ALL FLIGHTS ARE SUBJECT TO AIR TRAFFIC CONTROL SERVICE AND ARE SEPARATED FROM EACH OTHER.

AIRSPACE CLASSIFICATION

- **CLASS C:**

**IFR AND VFR FLIGHTS ARE PERMITTED,
ALL FLIGHTS ARE SUBJECT TO AIR
TRAFFIC CONTROL SERVICE AND IFR
FLIGHTS ARE SEPARATED FROM
OTHER IFR FLIGHTS AND FROM VFR
FLIGHTS.**

**VFR FLIGHTS ARE SEPARATED FROM IFR
FLIGHTS AND RECEIVE TRAFFIC INFO
ABOUT OTHER VFR TRAFFIC.**

AIRSPACE CLASSIFICATION

- **CLASS D** (NOT IN USE IN MUSCAT FIR)
**SAME LIKE CLASS C BUT HERE VFR FLIGHTS
RECEIVES TRAFFIC INFO ABOUT ALL OTHER
TRAFFIC.**
- **CLASS E** (NOT IN USE IN MUSCAT FIR)
**IFR AND VFR FLIGHTS ARE PERMITTED, IFR
FLIGHTS ARE SUBJECT TO AIR TRAFFIC
CONTROL SERVICE AND ARE SEPARATED
FROM OTHER IFR FLIGHTS. ALL FLIGHTS
RECEIVE TRAFFIC INFO AS FAR AS
PRACTICABLE.**

AIRSPACE CLASSIFICATION

- **CLASS F:**
IFR AND VFR FLIGHTS ARE PERMITTED.
ALL FLIGHTS RECEIVE AN AIR TRAFFIC
ADVISORY SERVICE.
- **CLASS G: (NOT IN USE IN MUSCAT FIR)**
IFR AND VFR FLIGHTS AND RECEIVE FLIGHT
INFO SERVICE IF REQUESTED.

Class	Type of Flight	Separation Provided	Service Provided	VMC Visibility and Distance from Cloud Minima	Speed Limitation	Radio Communication Requirement	ATC Clearance
A	IFR only	All aircraft	Air traffic control service	Not applicable	Not applicable	Continuous two-way	Yes
B	IFR	All aircraft	Air traffic control service	Not applicable	Not applicable	Continuous two-way	Yes
	VFR	All aircraft	Air traffic control service	8 km at and above 3050m (10000 FT) AMSL 5 km below 3050m (10000 FT) AMSL Clear of clouds	Not applicable	Continuous two-way	Yes

Class	Type of Flight	Separation Provided	Service Provided	VMC Visibility and Distance from Cloud Minima	Speed Limitation	Radio Communication Requirement	ATC Clearance
C	IFR	IFR from IFR IFR from VFR	Air traffic control service	Not applicable	250 KT IAS below 3050m (10000 FT) AMSL, unless advised by ATC.	Continuous two-way	Yes
	VFR	VFR from IFR	1) Air traffic control service for separation from IFR; 2) VFR /IFR traffic information (and traffic avoidance advice on request)	8 km at and above 3050m (10000 FT) AMSL 5 km below 3050m (10000 FT) AMSL 1500m horizontal; 300m vertical distance from cloud	250 KT IAS below 3050m (10000 FT) AMSL	Continuous two-way	Yes

Class	Type of Flight	Separation Provided	Service Provided	VMC Visibility and Distance from Cloud Minima	Speed Limitation	Radio Communication Requirement	ATC Clearance
D	IFR	IFR from IFR	Air traffic control service including traffic information about VFR flights (and traffic avoidance advice on request)	Not applicable	250 KT IAS below 3050m (10000 FT) AMSL	Continuous two-way	Yes
	VFR	Nil	Traffic information between VFR and IFR flights (and traffic avoidance advice on request)	8 km at and above 3050m (10000 FT) AMSL 5 km below 3050m (10000 FT) AMSL 1500m horizontal; 300m vertical distance from cloud	250 KT IAS below 3050m (10000 FT) AMSL	Continuous two-way	Yes

Class	Type of Flight	Separation Provided	Service Provided	VMC Visibility and Distance from Cloud Minima	Speed Limitation	Radio Communication Requirement	ATC Clearance
E	IFR	IFR from IFR	Air traffic control service and traffic information about VFR flights as far as practical	Not applicable	250 KT IAS below 3050m (10000 FT) AMSL	Continuous two-way	Yes
	VFR	Nil	Traffic information as far as practical	8 km at and above 3050m (10000 FT) AMSL 5 km below 3050m (10000 FT) AMSL 1500m horizontal; 300m vertical distance from cloud	250 KT IAS below 3050m (10000 FT) AMSL	No	No

Class	Type of Flight	Separation Provided	Service Provided	VMC Visibility and Distance from Cloud Minima	Speed Limitation	Radio Communication Requirement	ATC Clearance
F	IFR	IFR from IFR as far as practical	Air traffic advisory service; flight information service, radar advisory service.	Not applicable	250 KT IAS below 3050m (10000 FT) AMSL	Continuous two-way	No
	VFR	Nil	Flight information service, radar advisory service.	8 km at and above 3050m (10000 FT) AMSL 5 km below 3050m (10000 FT) AMSL 1500m horizontal; 300m vertical distance from cloud At and below 900m AMSL or 300m above terrain which ever is higher . 5 km (2) , clear of cloud and in sight of ground or water	250 KT IAS below 3050m (10000 FT) AMSL	No	No

Class	Type of Flight	Separation Provided	Service Provided	VMC Visibility and Distance from Cloud Minima	Speed Limitation	Radio Communication Requirement	ATC Clearance
G	IFR	Nil	Flight information service	Not applicable	250 KT IAS below 3050m (10000 FT) AMSL	Continuous two-way	No
	VFR	Nil	Flight information service	8 km at and above 3050m (10000 FT) AMSL 5 km below 3050m (10000 FT) AMSL 1500m horizontal; 300m vertical distance from cloud At and below 900m AMSL or 300m above terrain whichever is higher - 5 km, clear of cloud and in sight of ground or water	250 KT IAS below 3050m (10000 FT) AMSL	No	No

AIRSPACES USED IN OMAN

- A
- C
- F

VMC VISIBILITY AND DISTANCE FROM CLOUD MINIMA .

- 8 km AT AND ABOVE 10000`.
- 5 km BELOW 10000`.
- 1500M HORIZONTAL
- 300 M VERTICAL DISTANCE FROM CLOUD.

ARCS & RADIALS

ARCS & RADIALS

- Intercepting and Maintaining a Radial
- Maintaining/Correcting an Arc
- Arc Interception from a Radial
- Radial Interception from an Arc

ARCS & RADIALS

THE AIMS

ARCS & RADIALS

INTERCEPTING & MAINTAINING A RADIAL

- a. Using VOR needle
- b. Using beam bar on HSI

ARCS & RADIALS

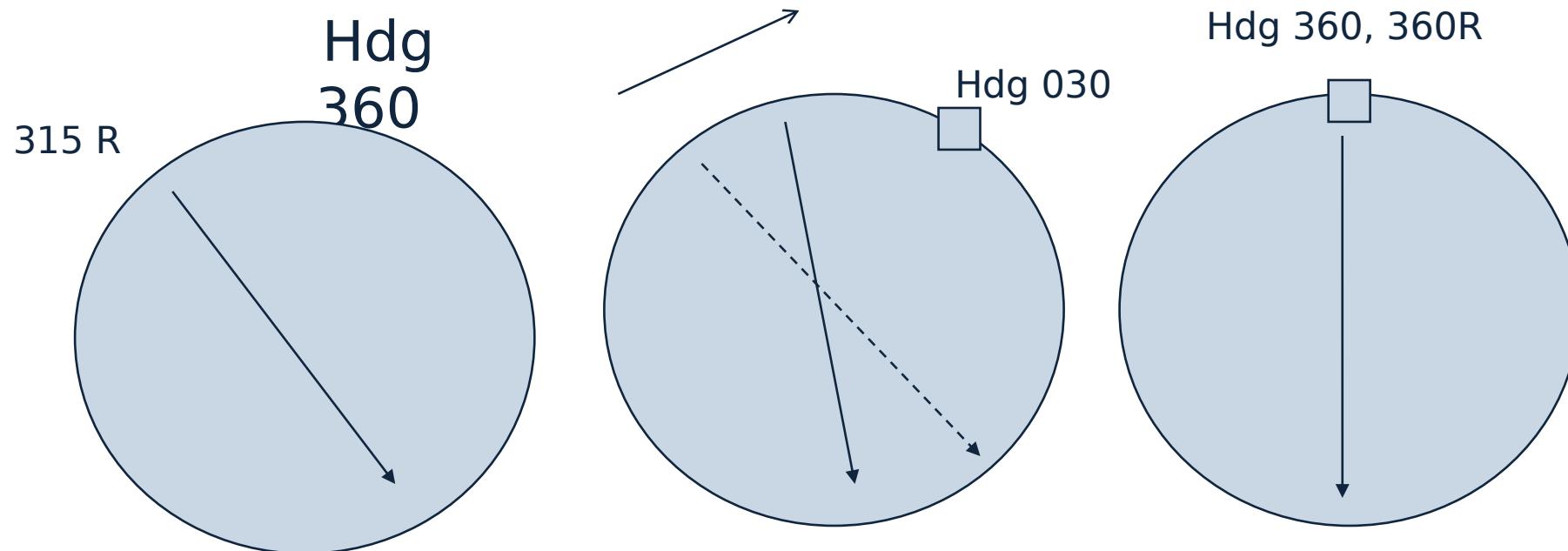
Using a VOR needle

- ‘Pull the tail, push the head’ of the needle to the required radial

ARCS & RADIALS

‘Pulling the tail’

On 315R, fly to intercept 360R

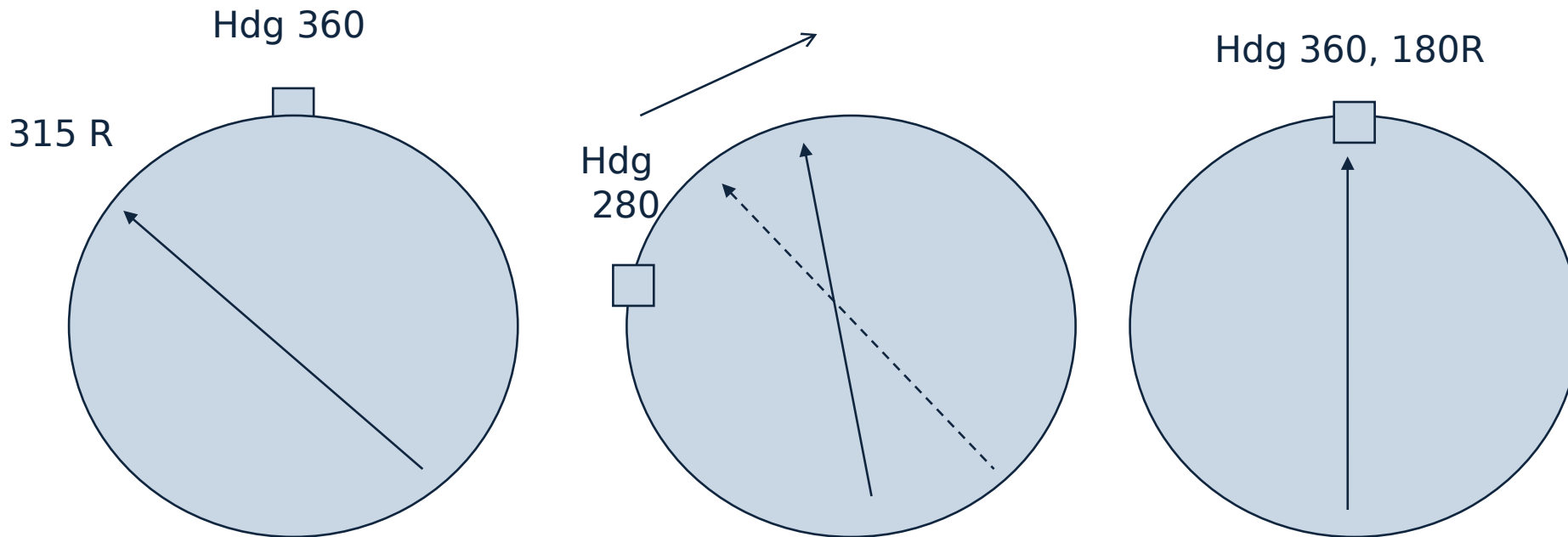


adjust hdg to offset drift!

ARCS & RADIALS

‘pushing the head’

- On 315(125R!), fly to intercept 360 inbound (180R!)

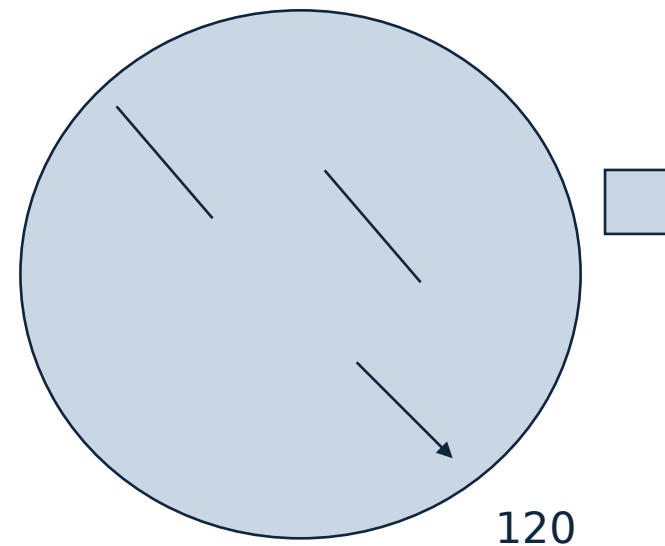
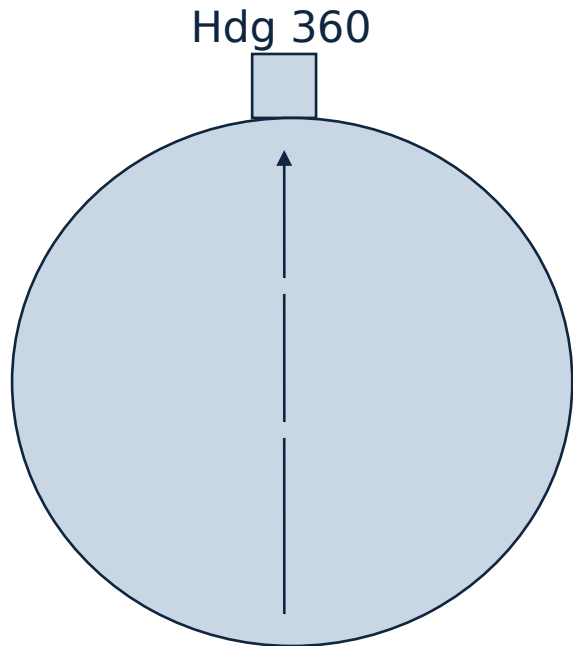


adjust hdg to offset drift!

ARCS & RADIALS

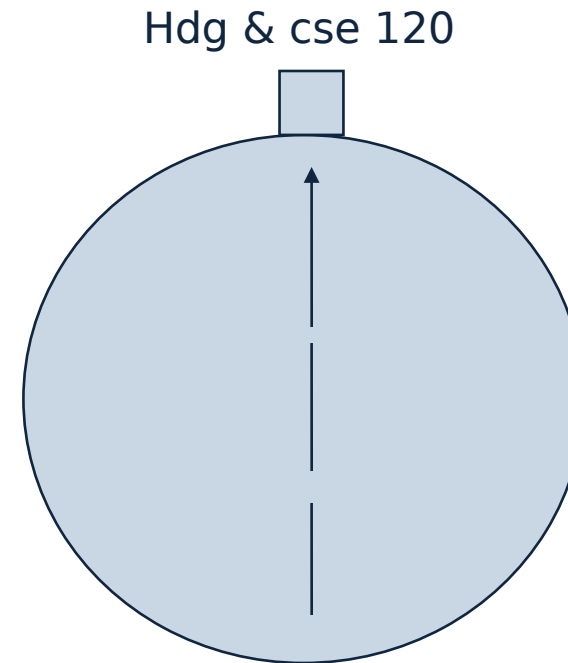
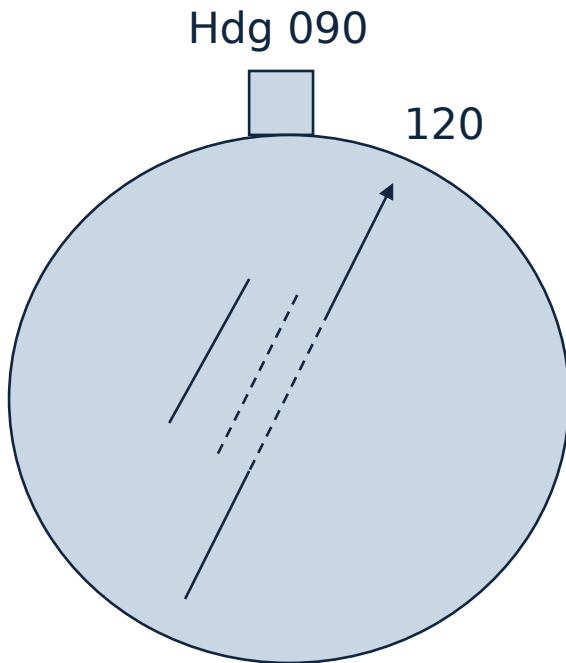
using an HSI

- **select required course (120) on HSI**



ARCS & RADIALS

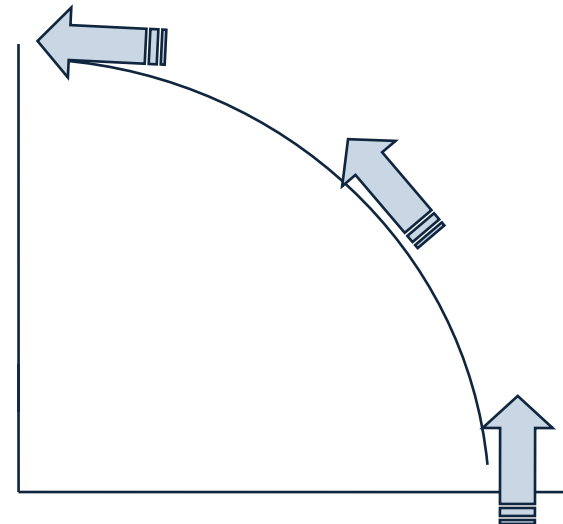
- **turn onto intercept hdg (090)**
- **establish on desired course**



ARCS & RADIALS

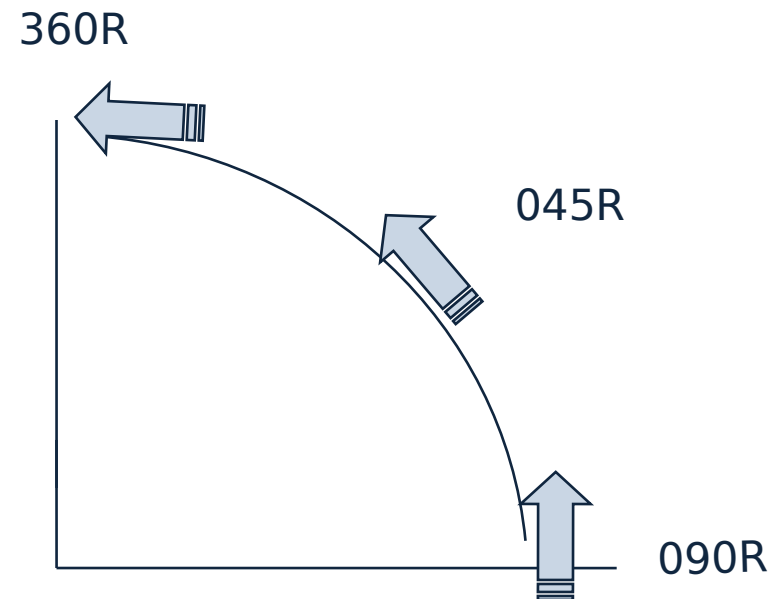
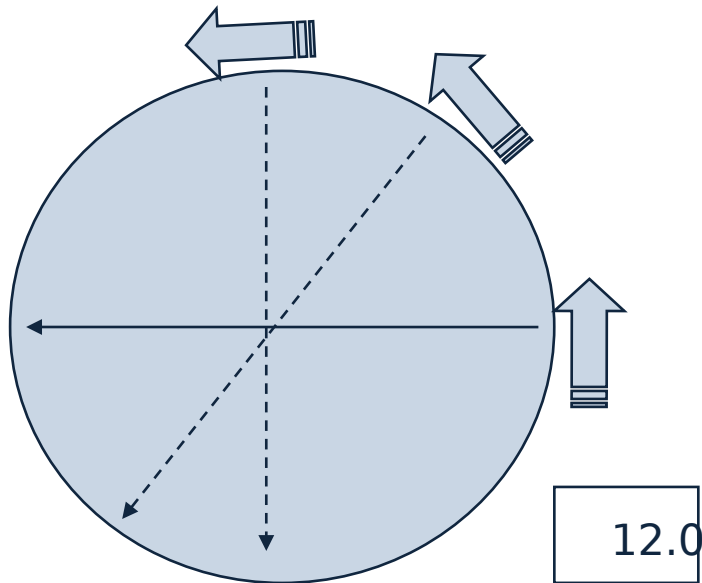
MAINTAINING AN ARC

- To maintain an arc the aircraft must be kept at 90° to the beacon



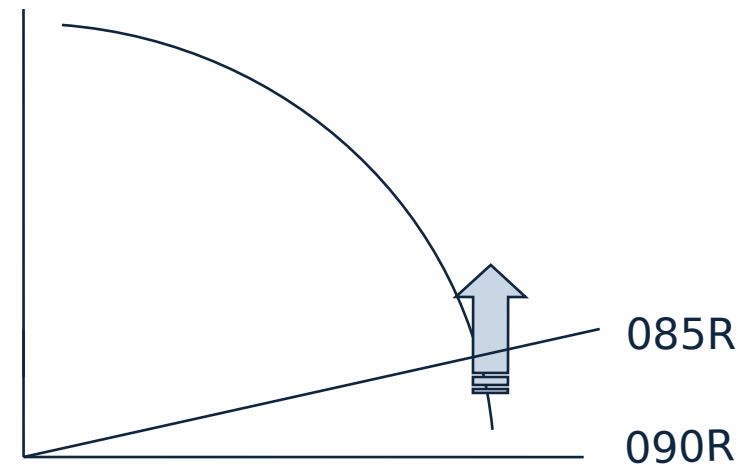
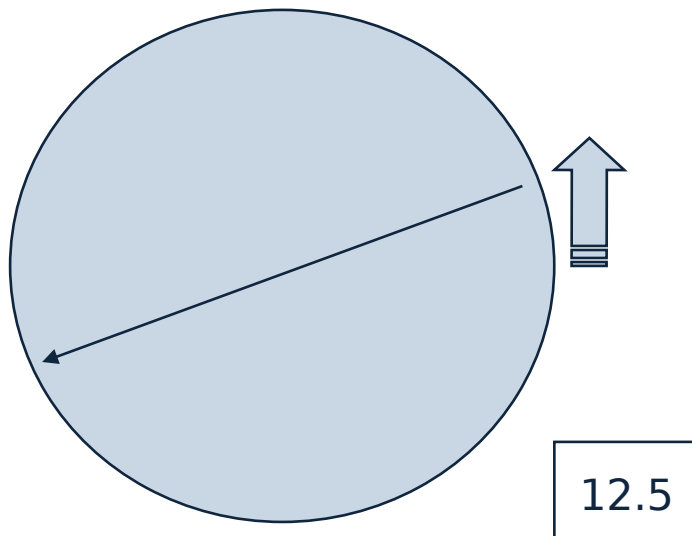
ARCS & RADIALS

- This is achieved by keeping the VOR needle at 90° to the aircraft heading on the compass



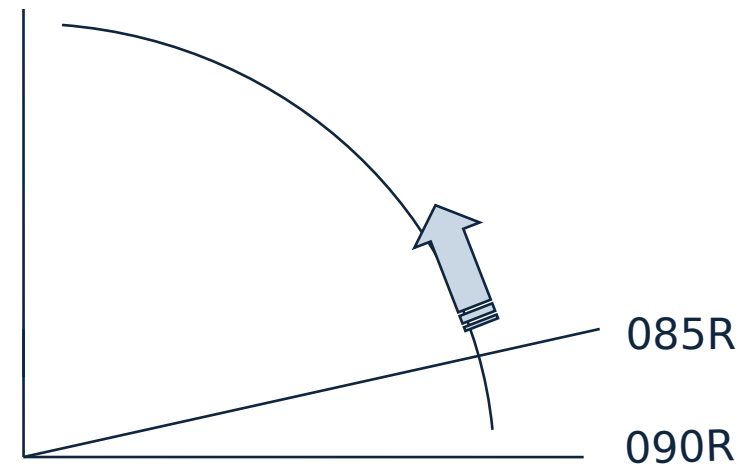
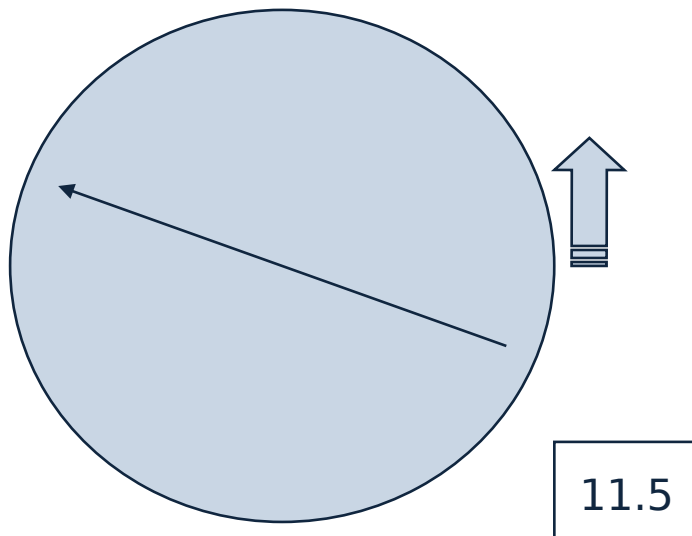
ARCS & RADIALS

- In practise, a heading is maintained until the needle 'drops' by 10° below the horizontal



ARCS & RADIALS

- The aircraft is then turned toward the beacon until the needle 'climbs' 10° above the horizontal



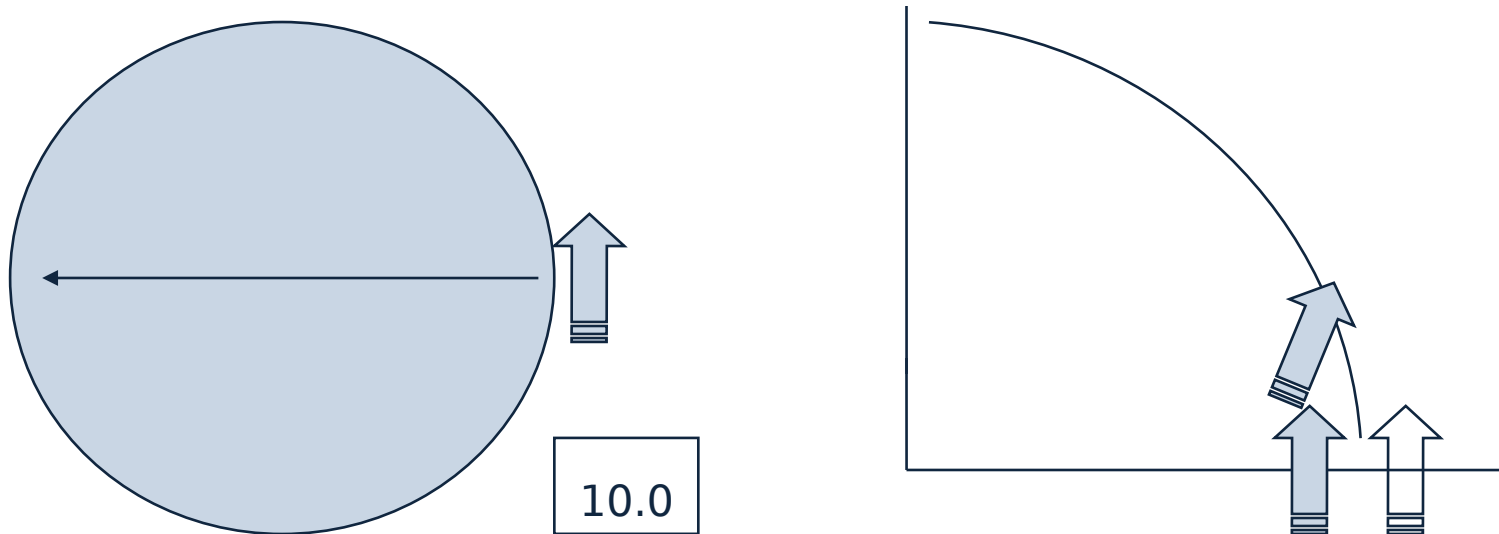
ARCS & RADIALS

- This heading is maintained as the needle passes through the horizontal & again 'drops' 10° below the horizontal.
- The aircraft is then turned again towards the beacon until the needle has climbed 10° above the horizontal.
- The process is repeated to maintain the arc!

ARCS & RADIALS

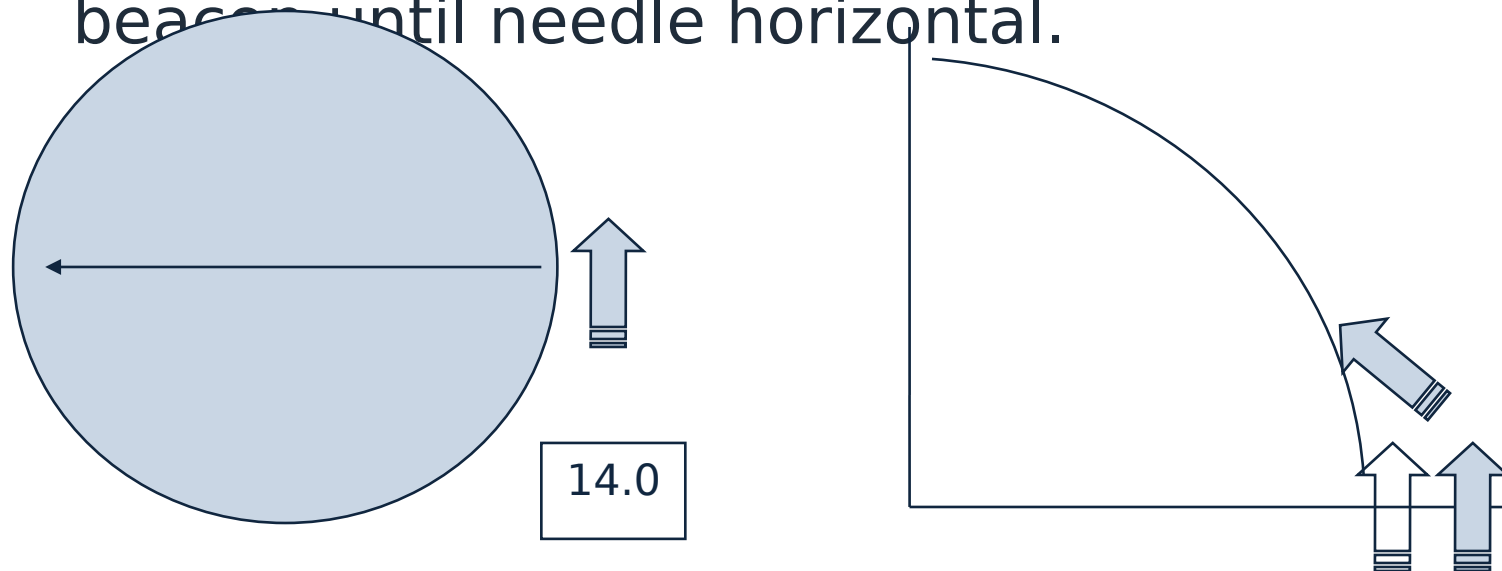
CORRECTING AN ARC

- a. When too close to the beacon
- turn aircraft away from beacon until correct distance achieved; turn aircraft back towards beacon until needle horizontal.



ARCS & RADIALS

- b. When too far out from the beacon
- turn aircraft towards beacon until correct distance achieved; turn aircraft away from beacon until needle horizontal.



ARCS & RADIALS

- ARC INTERCEPTION FROM A RADIAL
- Maintain radial until 0.5 mile from required arc
- At 0.5 mile from arc, turn aircraft until VOR needle at 90° to aircraft's heading
- Check actual distance from beacon:
 - if correct, maintain arc
 - if too close/too far out correct to required distance and then maintain!

ARCS & RADIALS

RADIAL INTERCEPTION FROM AN ARC

- maintain arc until 5° before reaching required radial
- turn aircraft onto desired heading for intercepting and maintaining required radial
- adjust aircraft heading as required to maintain radial!

ARCS & RADIALS

We have studied:

- Homing Principles
- Intercepting and Maintaining a Radial
- Maintaining/Correcting an Arc
- Arc Interception from a Radial
- Radial Interception from an Arc

ARCS & RADIALS

ANY
QUESTIONS?

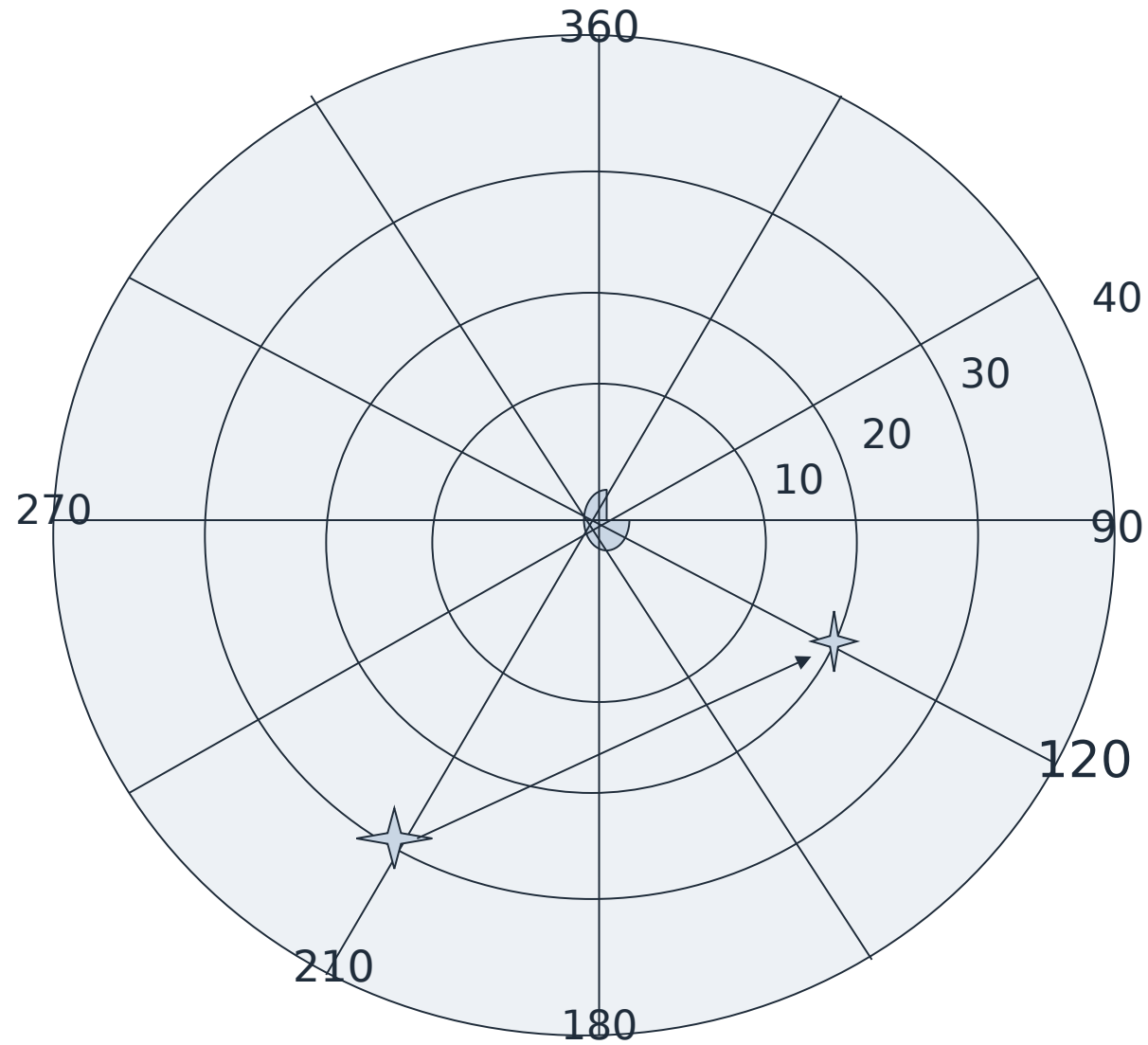
FIX TO FIX

AIM: TO FLY POINT TO POINT NAVIGATION
USING VOR-DME

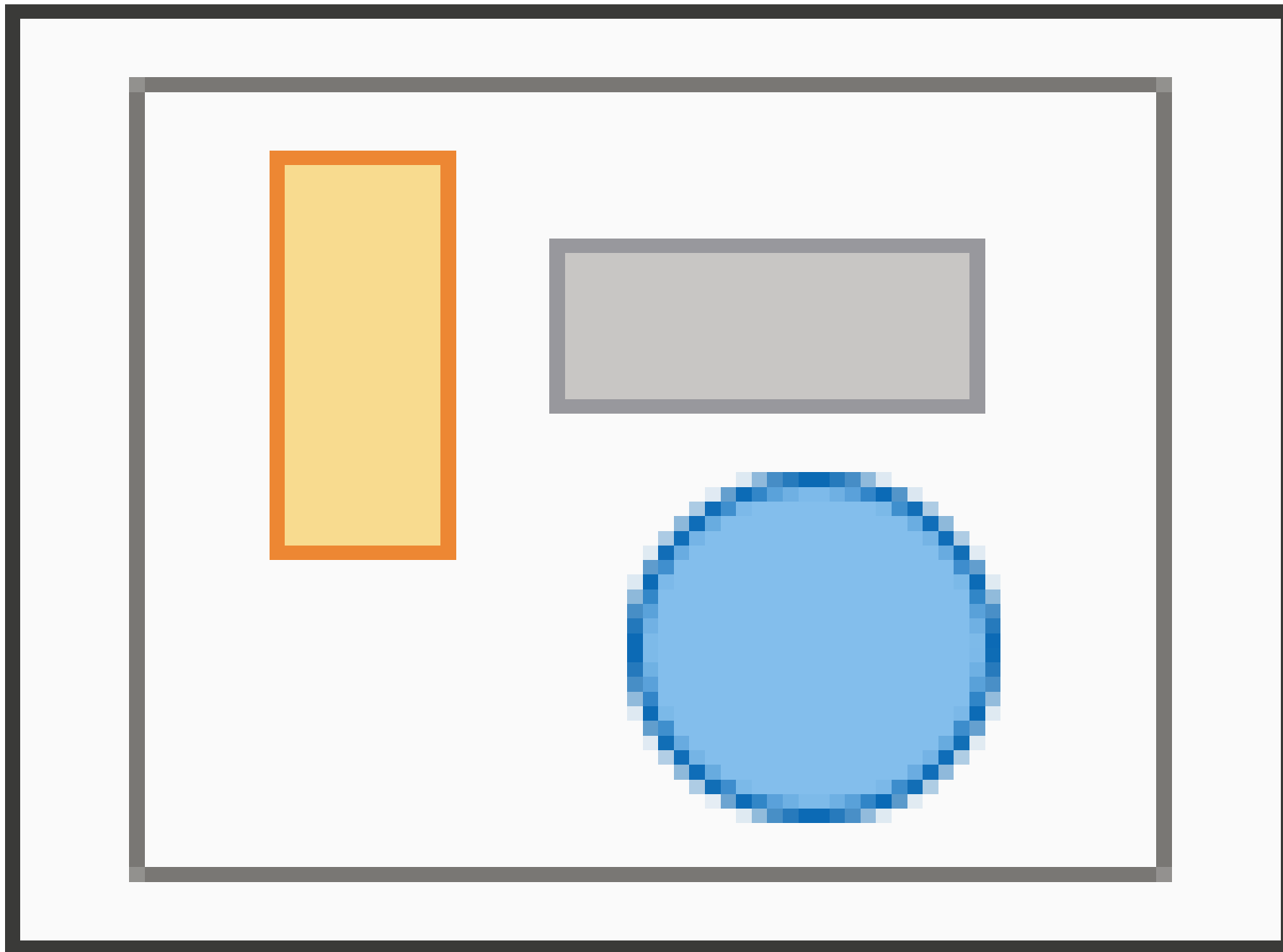
THER ARE TWO METHODES

- 1. KNEEPAD MAP**
- 2. HSI**

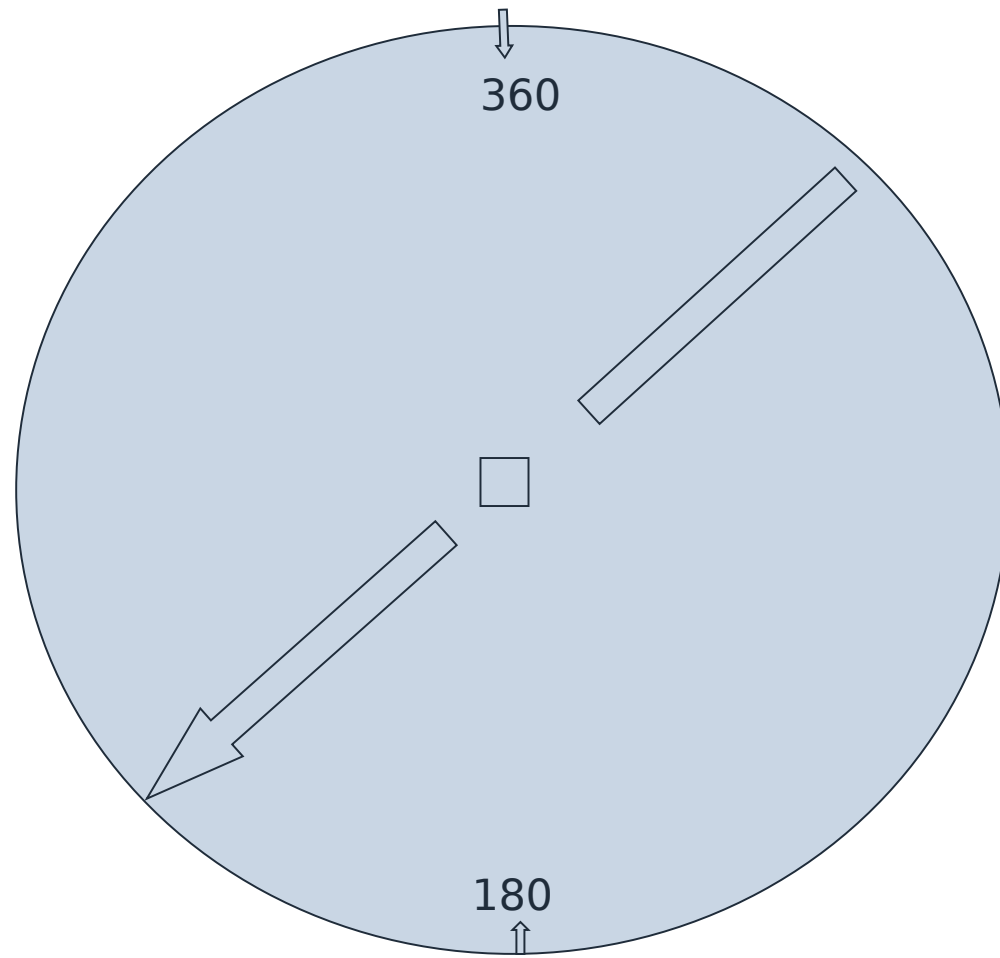
KNEEPAD MAP





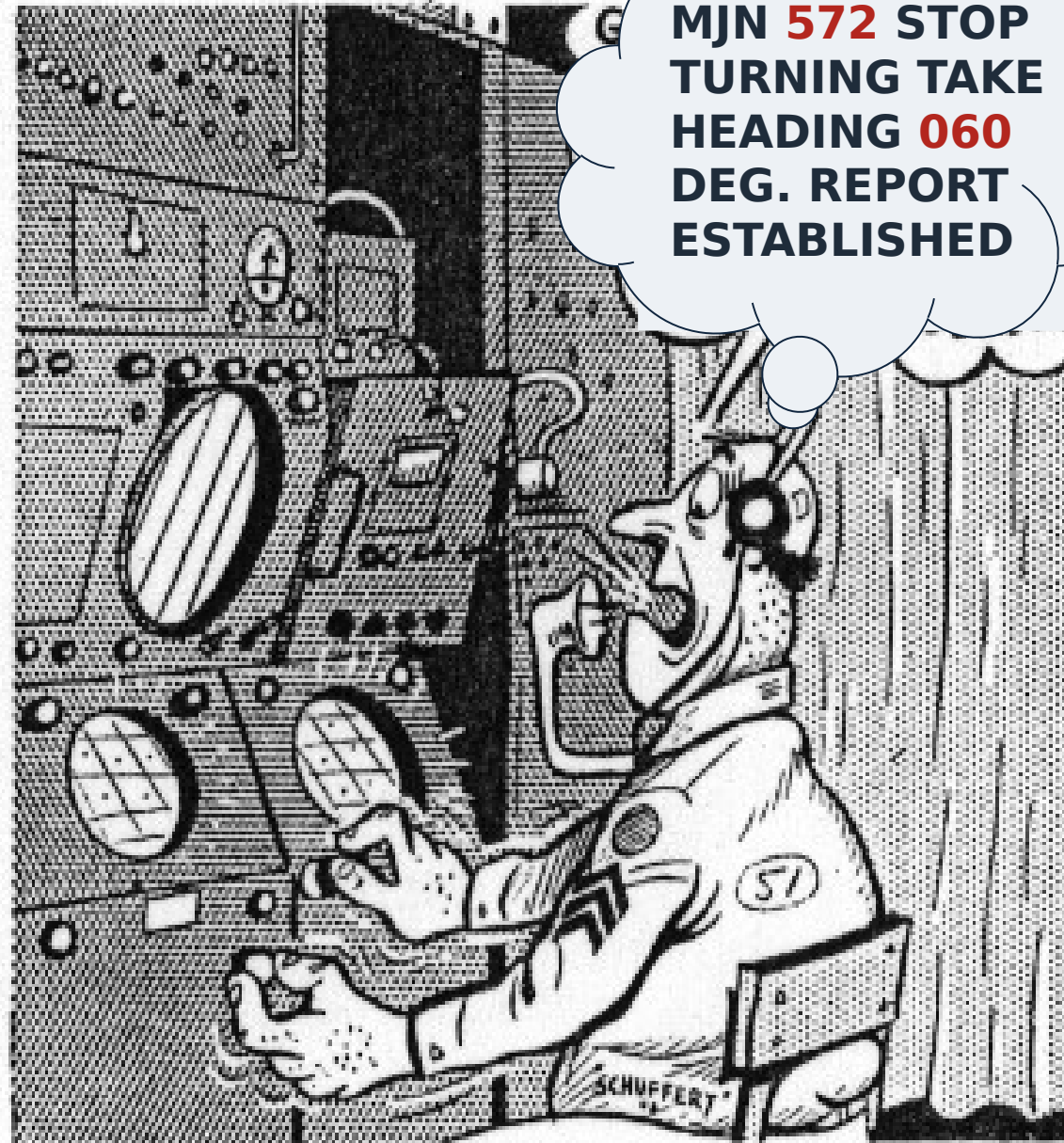


HSI



HOLDING PROCEDURE

**MJN 572 STOP
TURNING TAKE
HEADING 060
DEG. REPORT
ESTABLISHED**



HOLDING PROCEDURE

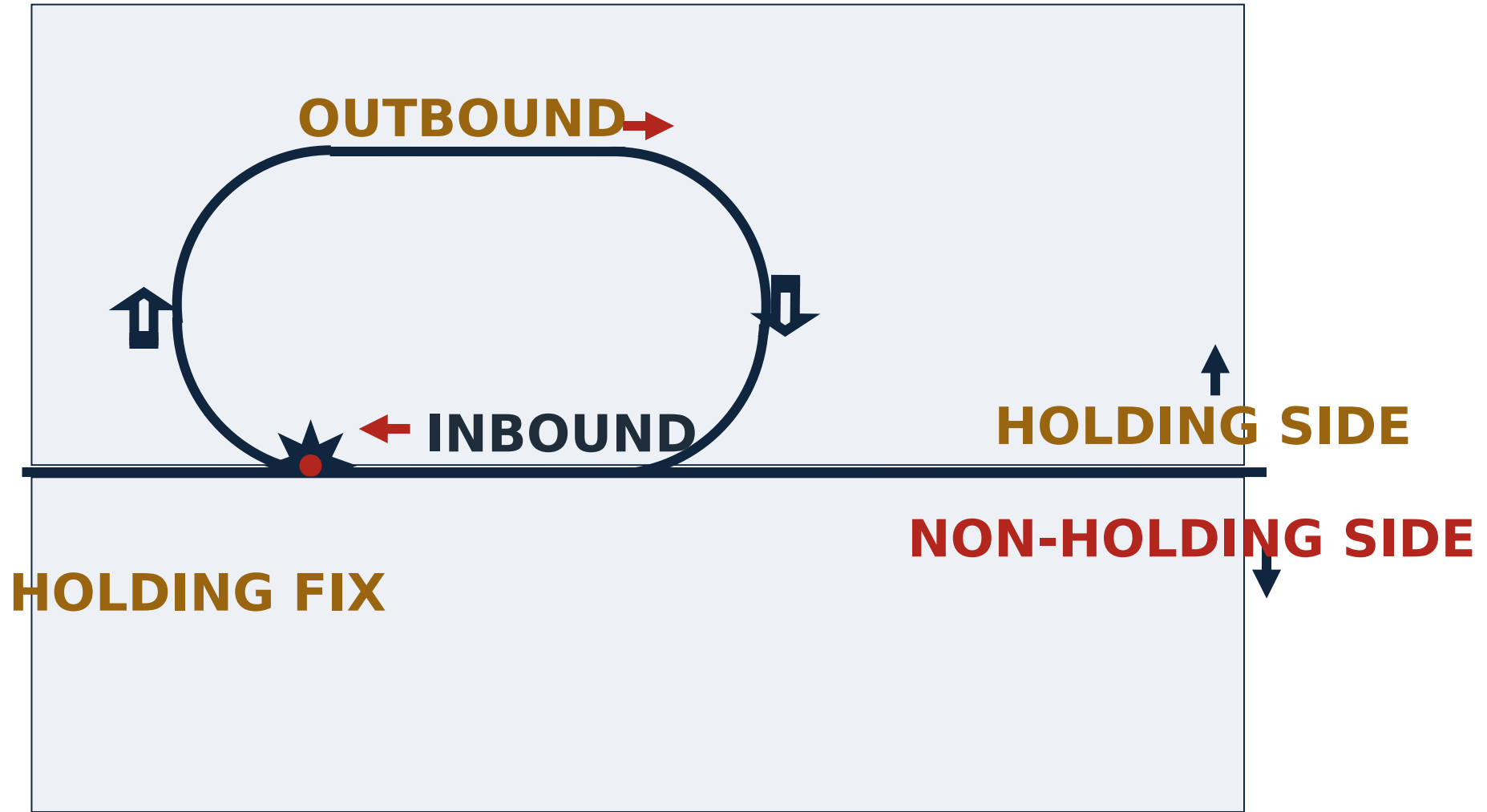
objectives

- TO RECOGNIZE THE 3 TYPES OF HOLDING PATTERN
- TO UNDERSTAND THE PROCEDURES TO BE USED WHEN HOLDING
- TO UNDERSTAND THE 3 TYPES OF ENTRY INTO A HOLDING PATTERN
- TO BE ABLE TO CALCULATE A MINIMUM HOLDING LEVEL
- TO KNOW YOUR AIRCRAFT'S HOLDING SPEED

HOLDING

**HOLDING PROCEDURE ORIENTATE THE A/C ON THE
CORRECT HEADING TO FLY THE APPROACH
PROCEDURE AND ALLOW ATC TO SEQUENCE
SEVERAL A/C INTO THE APPROACH PATTERN.**

HOLDING PROCEDURES



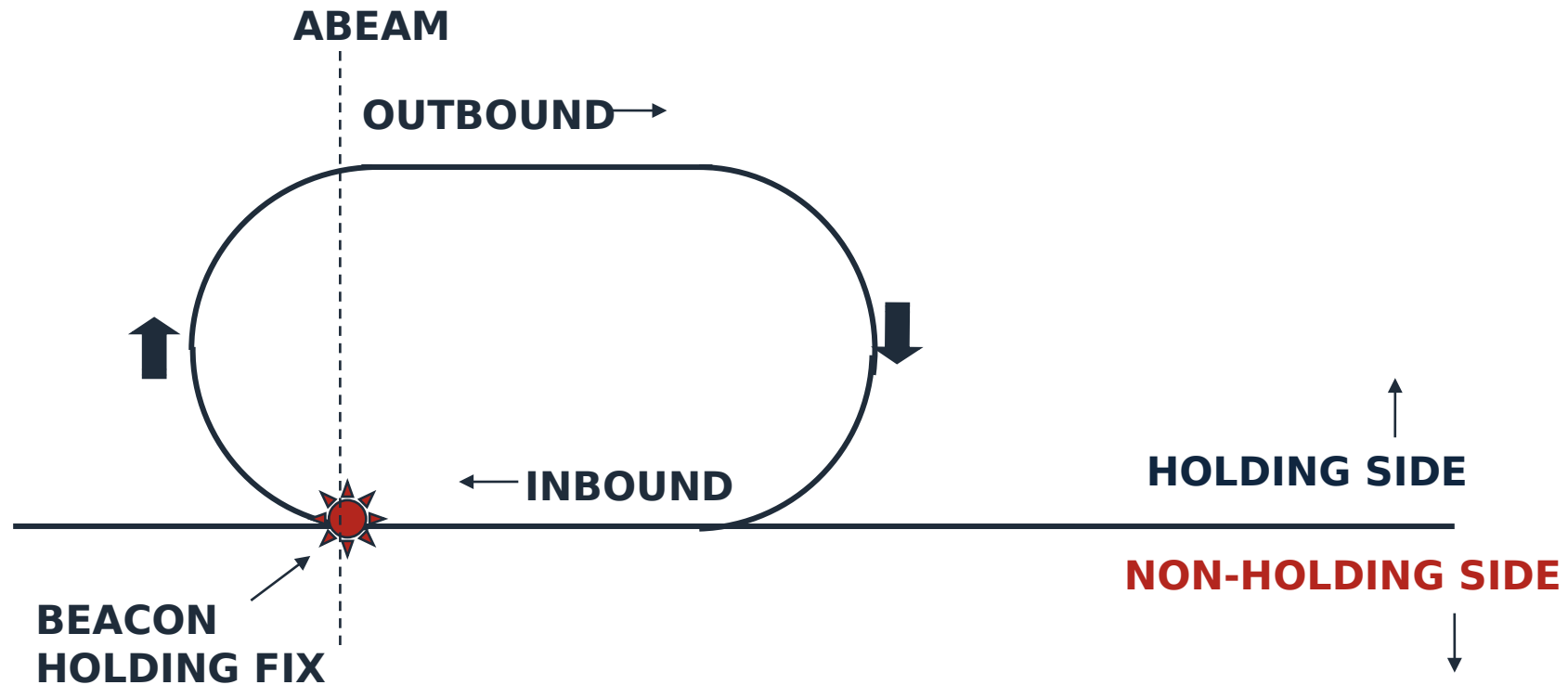


TYPES OF HOLDING PATTERN

- **HOLDING OVERHEAD A BEACON**
- **VOR/DME - HOLDING TOWARDS THE BEACON.**
- **VOR/DME - HOLDING AWAY FROM THE BEACON.**

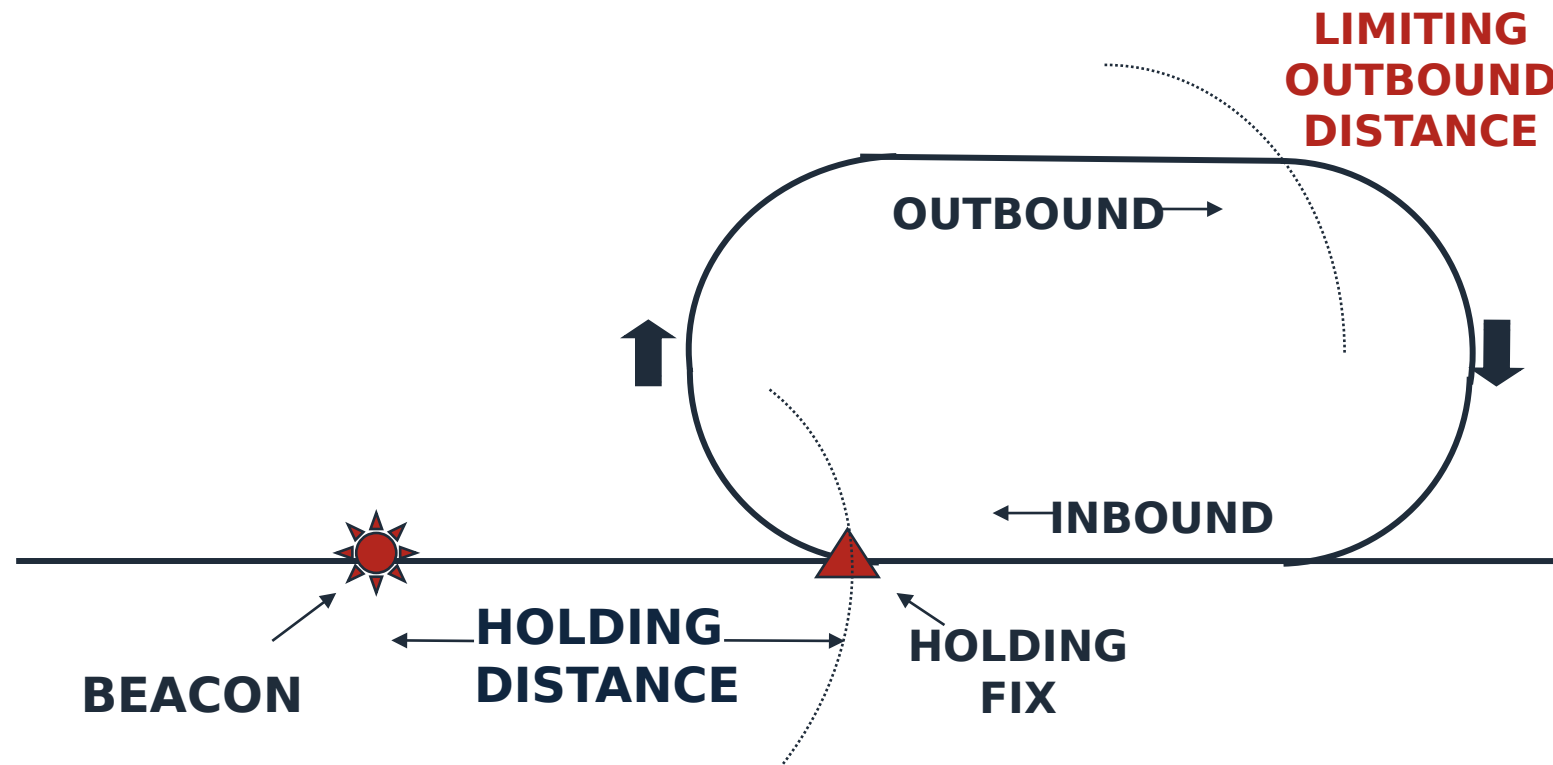
TYPES OF HOLDING PATTERN

1. HOLDING OVERHEAD A BEACON



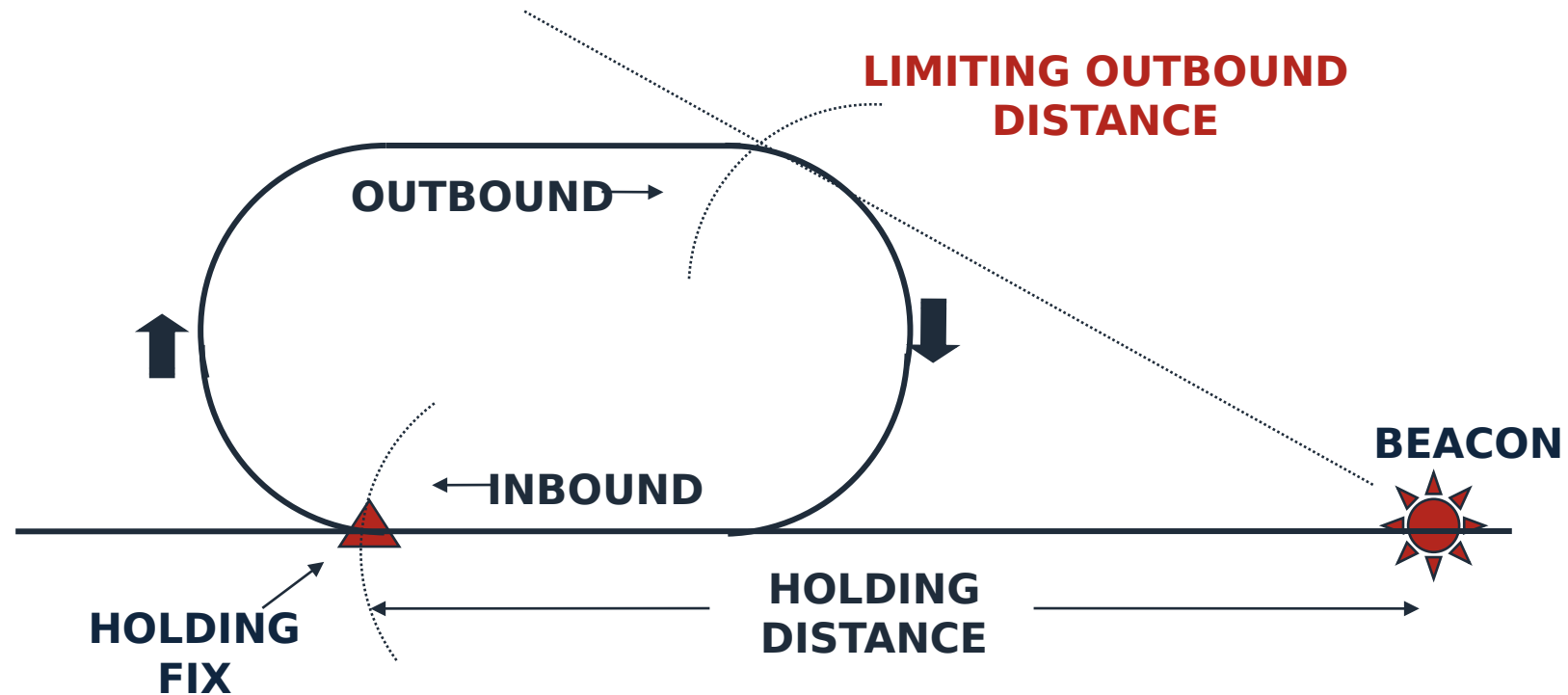
TYPES OF HOLDING PATTERN

2. VOR/DME, HOLDING TOWARDS THE BEACON



TYPES OF HOLDING PATTERN

3. VOR/DME, HOLDING AWAY FROM THE BEACON



HOLDING PROCEDURES

- **ALL TURNS TO BE RATE 1 (3°/SECOND) OR 25° AOB.**
- **ALL PROCEDURES DEPICT TRACKS, TRACKS SHOULD BE MAINTAINED BY MAKING ALLOWANCE FOR WIND BY APPLYING CORRECTIONS TO HEADINGS AND TIMINGS DURING ENTRY AND HOLDING PROCEDURES.**
- **OUTBOUND TIMING STARTS ABEAM THE FIX (IF ABEAM POSITION CANNOT BE DETERMINED, START TIMING UPON COMPLETION OF OUTBOUND TURN).**
- **IF THE OUTBOUND LEG LENGTH IS BASED ON A DME DISTANCE. THE OUTBOUND LEG ENDS AS SOON AS THE LIMITING DME DISTANCE IS ACHIEVED**

ENTRY

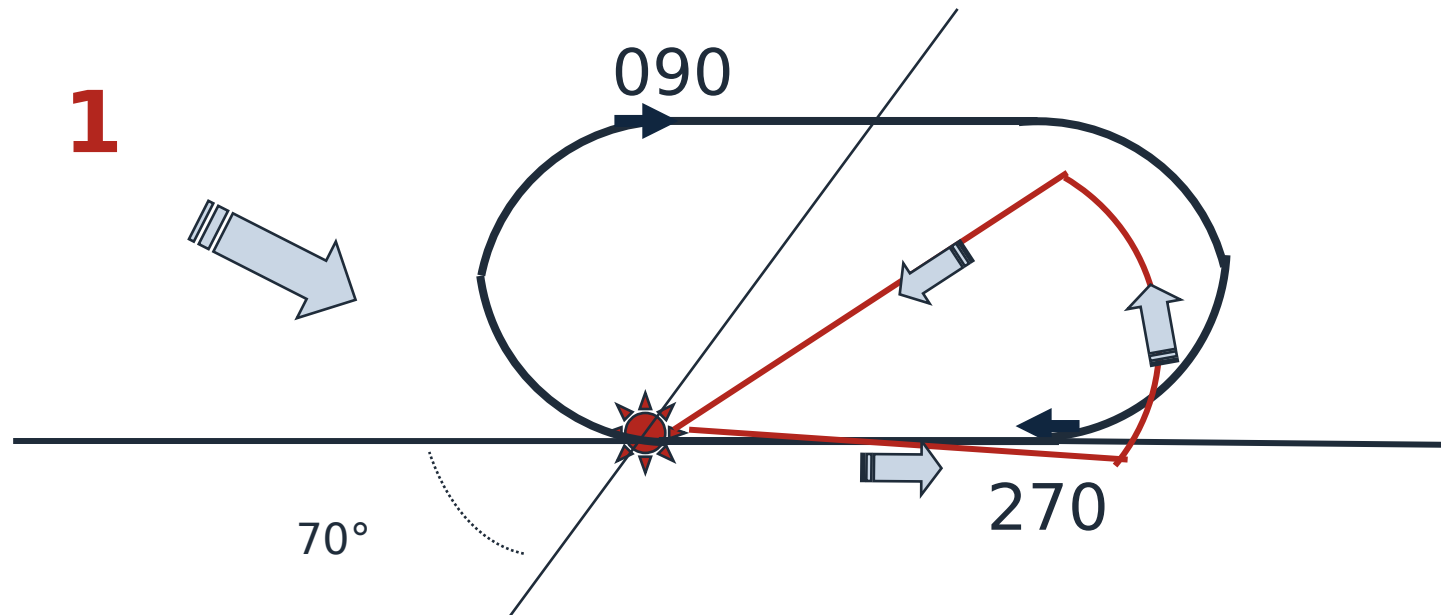
- **There are 3 entry sectors:**
 sector 1 (parallel entry)
 sector 2 (offset entry)
 sector 3 (direct entry)

ENTRY

SECTOR 1, (PARALLEL ENTRY)

FROM OVERHEAD FIX, A/C TURNED ONTO OUTBOUND HEADING (1 MINUTE, WIND ADJUSTED).

A/C TURNED ONTO HOLDING SIDE TO INTERCEPT INBOUND TRACK OR TO RETURN TO FIX.

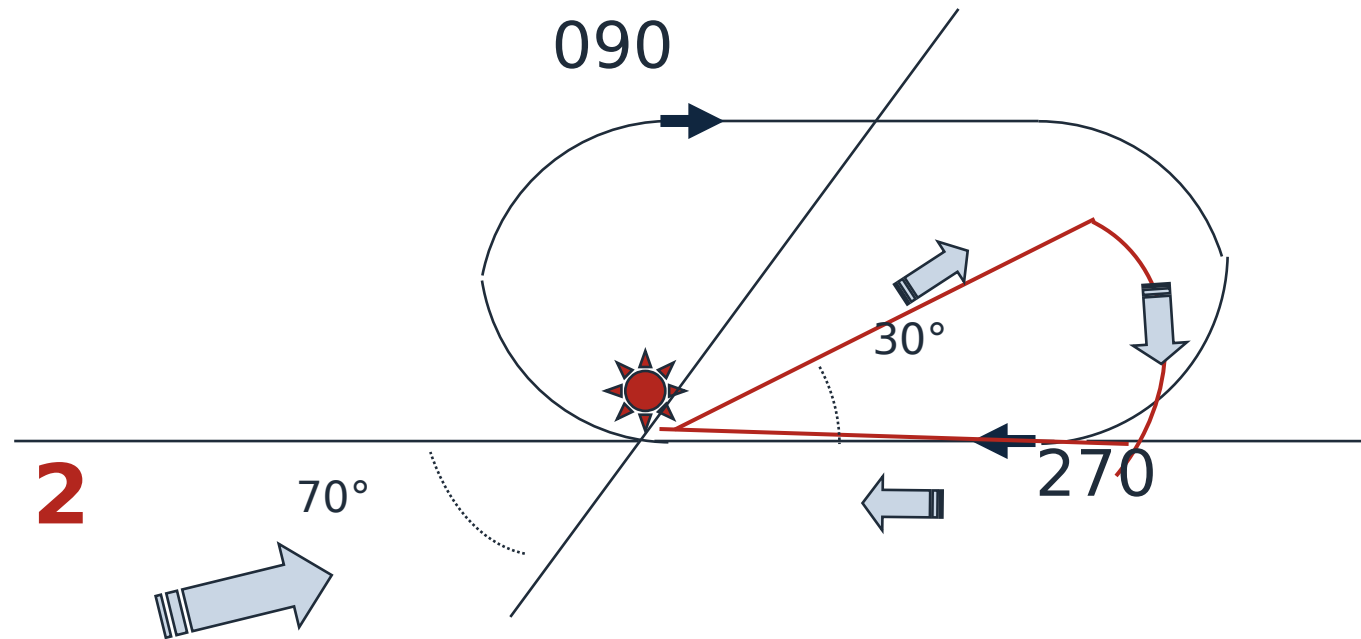


ENTRY

SECTOR 2, (OFFSET ENTRY)

FROM OVERHEAD THE FIX, A/C TURNED ONTO TRACK 30° OFFSET (1 MINUTE, WIND-ADJUSTED OR UNTIL LIMITING DME).

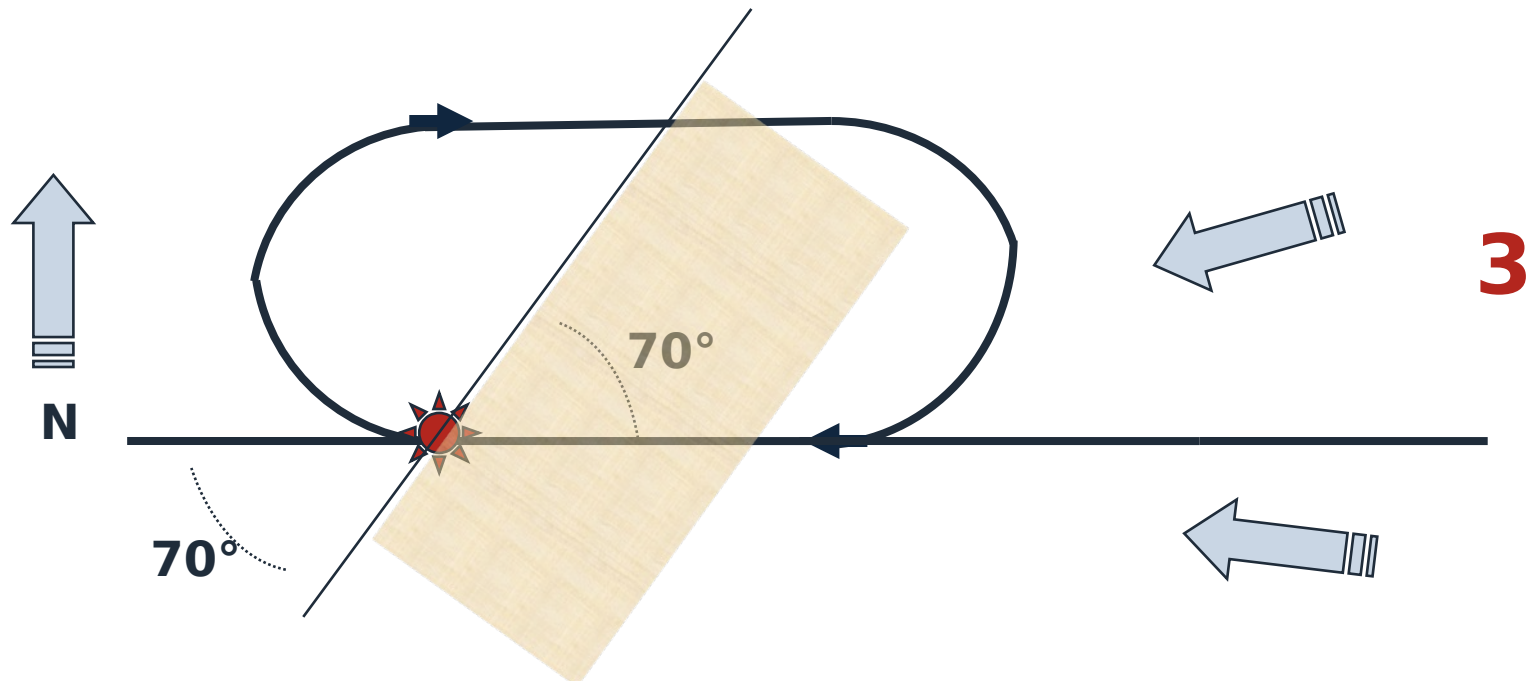
A/C TURNED TO INTERCEPT INBOUND TRACK.



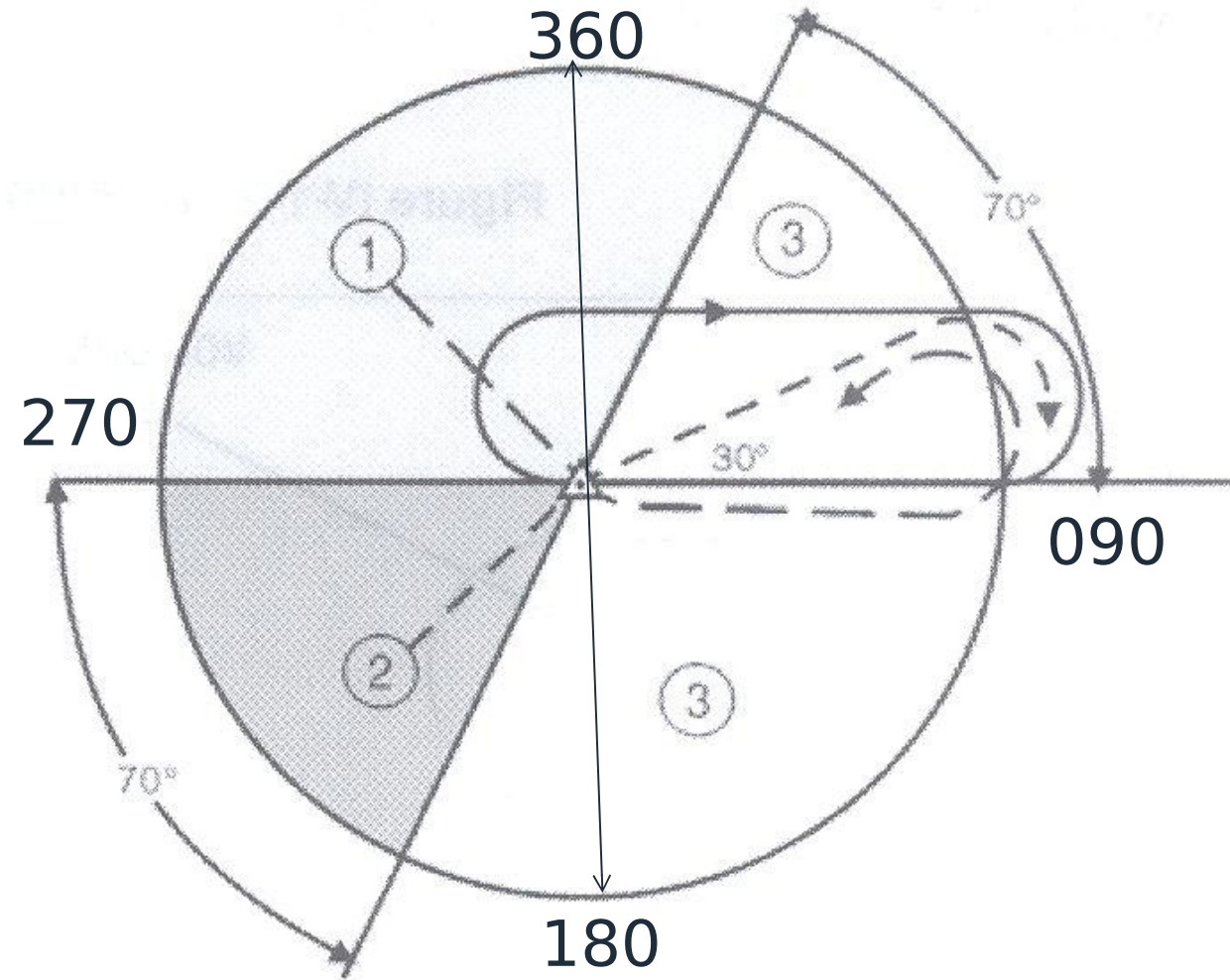
ENTRY

SECTOR 3, (DIRECT ENTRY)

HAVING REACHED THE FIX, A/C IS TURNED TO FOLLOW THE HOLDING PATTERN



THE THREE HOLDING SECTORS



ENTRY

- **The entry to be used into the holding pattern depends on the aircraft heading in relation to the 3 entry sectors**
- **For entry, there is a zone of flexibility of 5° on either side of the sector boundaries**

HOLDING PROCEDURES

Minimum Holding Level

- The minimum holding level provides **TERRAIN CLEARANCE** in the **holding area** (the airspace required for the entry and holding patterns) and the **buffer area** (an area extending 5nm beyond the boundary of the holding area)
- The minimum holding level provides
 - the holding area a clearance of 300m (984') above obstacles in the holding area
 - the buffer area a clearance above obstacles in the buffer area; the extent of the clearance reduces with the distance from the holding area

AIRCRAFT HOLDING

SPEEDS

AN AIRCRAFT'S HOLDING SPEED DEPENDS UPON WHETHER IT IS A JET OR TURBOPROP AIRCRAFT, AND THE HOLDING ALTITUDE.

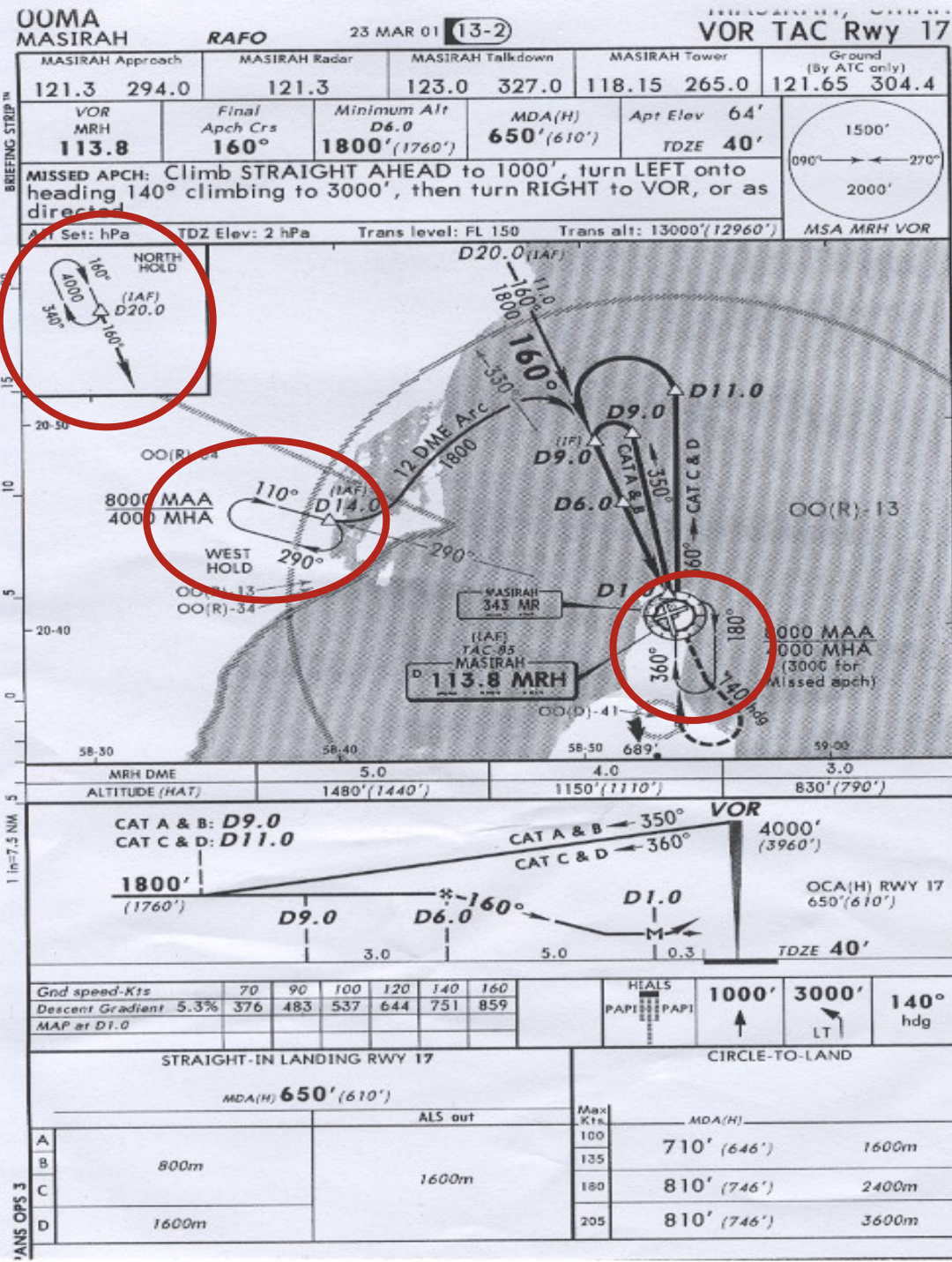
ALT/LEVEL	TURBOPROP	JET
UP TO 14,000'	170KTS	210 KTS
ABOVE 14,000'	175 KTS	240 KTS

HOLDING PROCEDURE

CONCLUSION

WE HAVE:

- ☛ **THE 3 TYPES OF HOLDING PATTERN**
- ☛ **THE PROCEDURES TO BE USED WHEN HOLDING**
- ☛ **THE 3 TYPES OF ENTRY INTO A HOLDING PATTERN**
- ☛ **THE CALCULATION OF A MINIMUM HOLDING LEVEL**
- ☛ **AIRCRAFT HOLDING SPEEDS**



HOLDING PROCEDURE

ANY
QUESTIONS?

TERMINAL APPROACH PLATES (TAPs)

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TERMINAL APPROACH PLATES (TAPs)

AIMS:

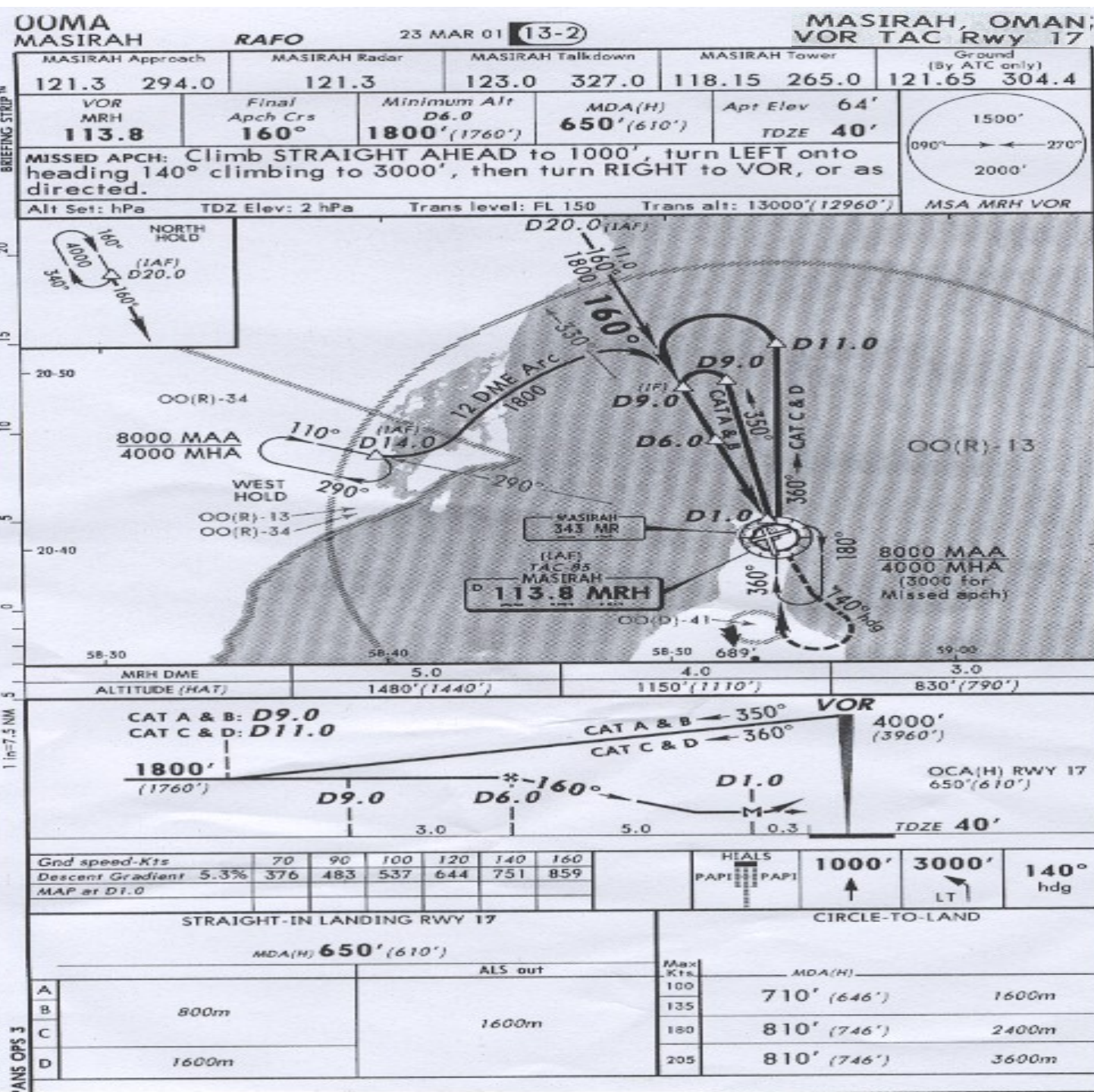
- TO UNDERSTAND THE ABBREVIATIONS USED ON TAPS
- TO UNDERSTAND THE SYMBOLS USED ON TAPS
- TO LEARN HOW TO INTERPRET A PRECISION & A NON-PRECISION APPROACH

TERMINAL APPROACH PLATES (TAPs)

ABBREVIATIONS & SYMBOLS

TAP STUDY

- MASIRAH VORTAC RWY 17



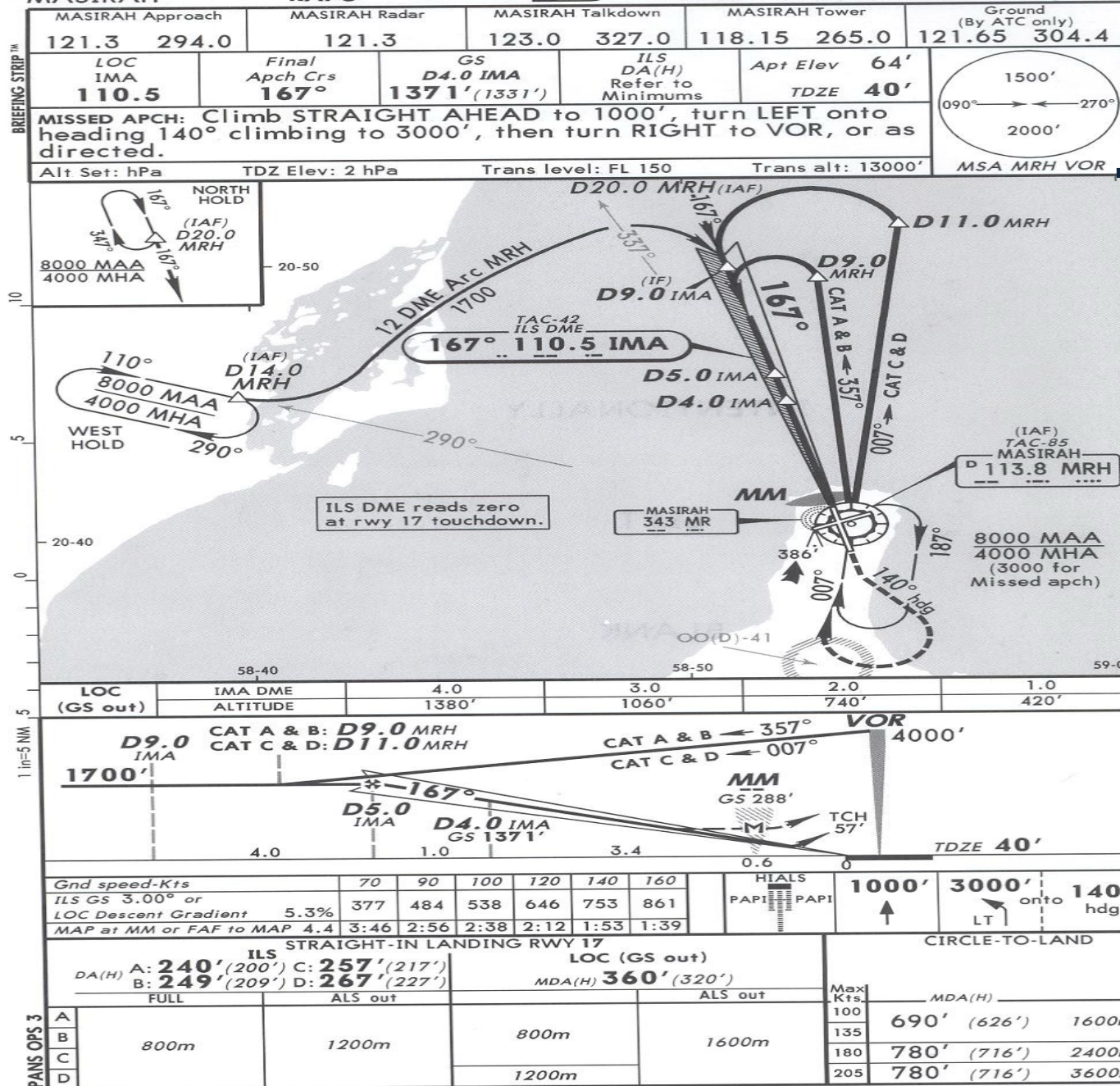
OOMA/MSH MASIRAH

RAFO

21 SEP 05

(11-1)

VOR TAC ILS DME Rwy 17



heading

Plan view

PROFILE
VIEW

LANDING
MINIMUMS

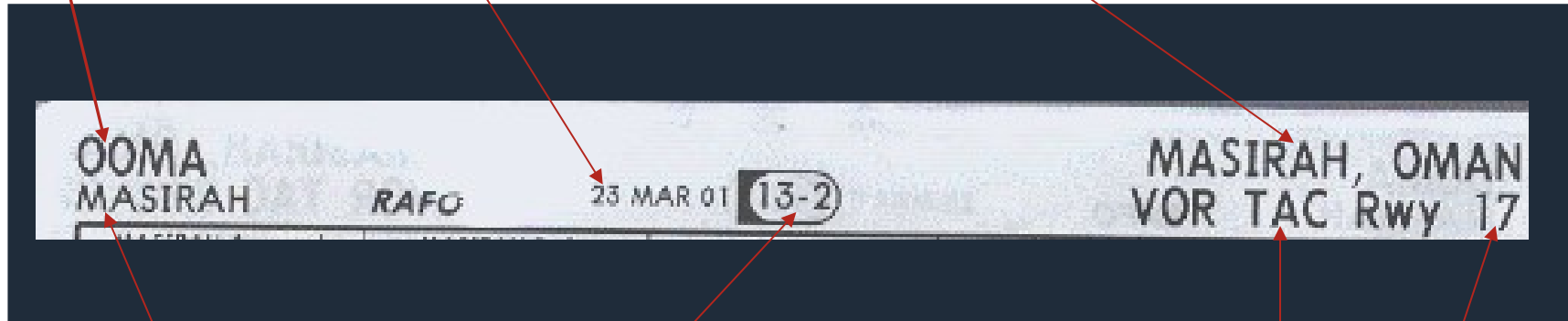
CHANGES: Profile. Note removed.

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**GEOGRAPHIC
NAME**

CHART DATE

GEOGRAPHICAL LOCATION



AIRPORT NAME

CART INDEX NUMBER

PROCEDURE IDENTIFICATION

DESIGNATED RUNWAY

UOMA MASIRAH		RAFO		23 MAR 01		13-2		MASIRAH, OMAN VOR TAC Rwy 17	
MASIRAH Approach		MASIRAH Radar		MASIRAH Talkdown		MASIRAH Tower		Ground (By ATC only)	
121.3 294.0		121.3		123.0 327.0		118.15 265.0		121.65 304.4	
VOR MRH 113.8		Final Apch Crs 160°		Minimum Alt D6.0 1800' (1760')		MDA(H) 650' (610')		Apt Elev 64' TDZE 40'	
MISSED APCH: Climb STRAIGHT AHEAD to 1000', turn LEFT onto heading 140° climbing to 3000', then turn RIGHT to VOR, or as directed.									
Alt Set: hPa		TDZ Elev: 2 hPa		Trans level: FL 150		Trans alt: 13000' (12960')		MSA MRH VOR	

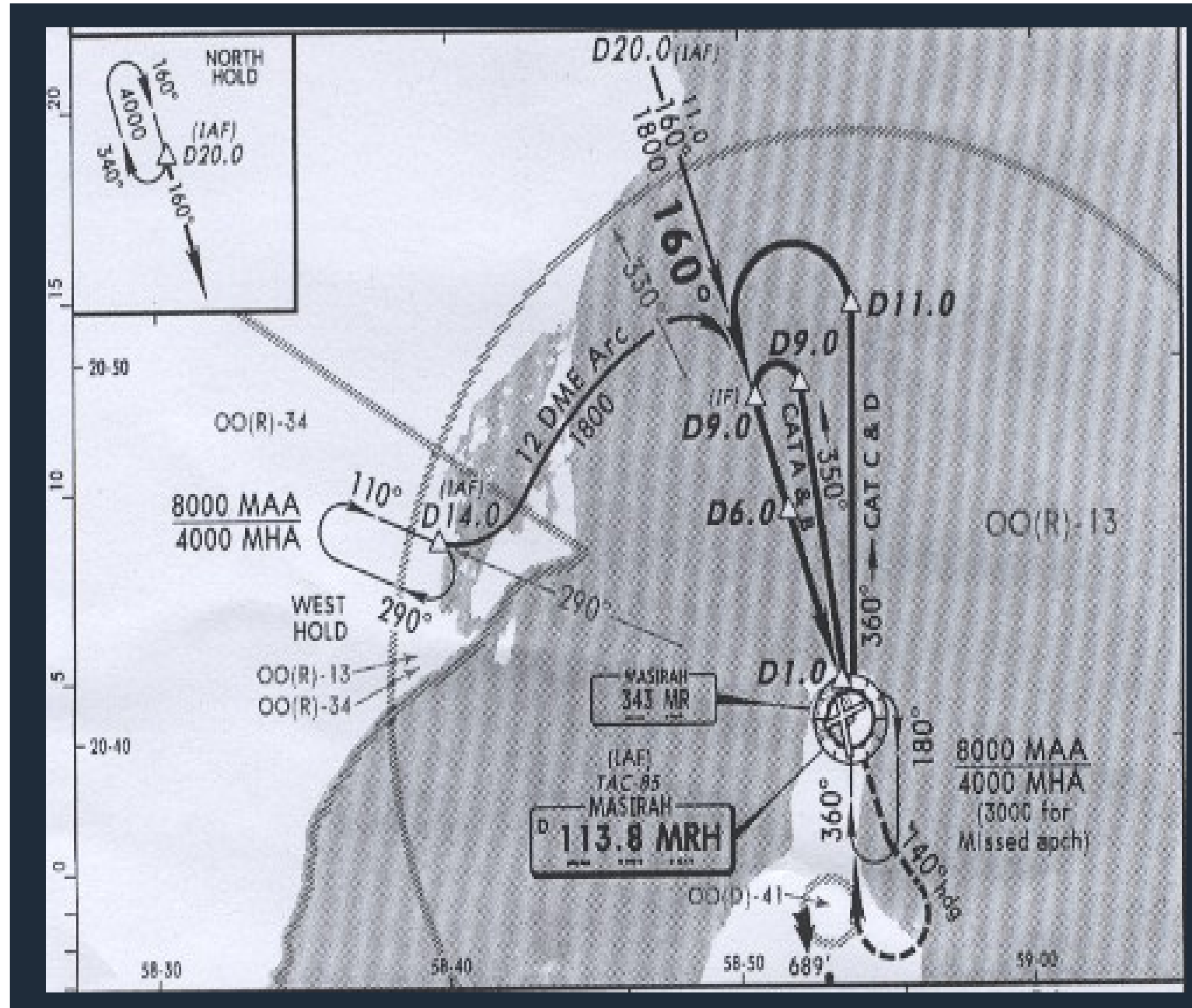
ALTIMETER SETTING

MSA Minimum Safe Altitude

T.L IS THE LOWEST LEVEL OF FLIGHT USING STANDARD ALTIMETER SETTING (OF 1013.25 MILLIBARS).

T.A IS THE ALTITUDE AT AND BELOW WICH LOCAL PRESSURE SETTING MUST BE USE (QNH).

APPROACH PLAN VIEW



Aircraft Category	V _{at}	Range of Speeds for Initial Approach	Range of Final Approach Speeds	Max speeds for Visual Maneuvering (Circling)	Max speeds for Missed Approach	
					Intermediate	Final
A	<91	90/150(110*)	70/100	100	100	110
B	91/120	120/180(140*)	85/130	135	130	150
C	121/140	160/240	115/160	180	160	240
D	141/165	185/250	130/185	205	185	265
E	166/210	185/250	155/230	240	230	275

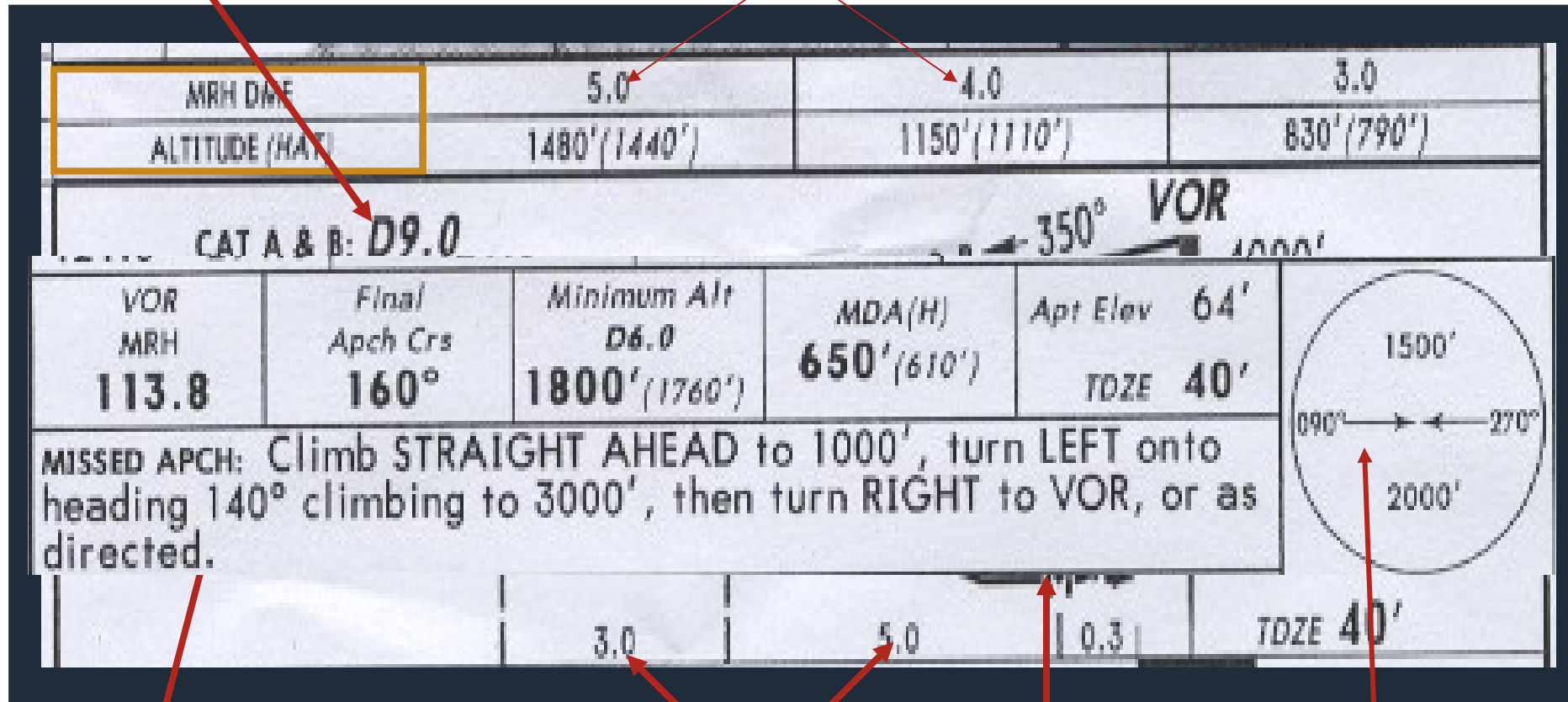
V_{at}—Speed at threshold based on 1.3 times stall speed in the landing configuration at maximum certificated landing mass.

*Maximum speed for reversal and racetrack procedures.

Category E contains only certain Military Aircraft and is not included on Jeppesen Approach Charts.

PROCEDURE TURN

DME FIXES



MINIMUM ALTITUDE
AND HIGHT ABOVE
TDZE

NM DISTANCE
BETWEEN
FIXES

MAP

OBSTACLE
CLEARANCE
ALTITUDE

Gnd speed-Kts		70	90	100	120	140	160		HEALS PAPI PAPI	1000'	3000'	140° hdg
Descent Gradient 5.3%		376	483	537	644	751	859			↑	↖ LT	
MAP at D1.0												
STRAIGHT-IN LANDING RWY 17								CIRCLE-TO-LAND				
MDA(H) 650' (610')												
				ALS out				Max Kts	MDA(H)			
A	800m			1600m				100	710' (646')		1600m	
B								135				
C								180	810' (746')		2400m	
D	1600m							205	810' (746')		3600m	

INSTRUMENT APPROACH PROCEDURES

OBJECTIVES:

- To identify the 4 segments of an instrument approach.
- To examine the differences between a precision and a non-precision approach.
- To study the circling approach.

INSTRUMENT APPROACH PROCEDURES

Segments of an Instrument Approach

There are 4 separate segments to an instrument approach:

- Initial
- Intermediate
- Final
- Missed Approach

INSTRUMENT APPROACH PROCEDURES

Arrival Segment

Standard Instrument Arrival (STAR)

May be replaced by radar vectoring

INSTRUMENT APPROACH PROCEDURES

Initial Approach Segment

- From Initial Approach Fix (IAF) to Intermediate Fix (IF).
- Aircraft has departed the en-route structure and is maneuvered to enter the intermediate approach sector.

INSTRUMENT APPROACH PROCEDURES

Intermediate Approach Segment

- From Intermediate Fix (IF) to Final Approach Fix (FAF).
- Aircraft's speed and configuration adjusted to prepare for the final approach.

INSTRUMENT APPROACH PROCEDURES

Final Approach Segment

- From Final Approach Fix (FAF) to Missed Approach Point (MAP).
- Aircraft aligned and descended for landing.
- Final approach may be made to runway for straight-in landing or to an airfield for visual, circling, maneuvers to land.

INSTRUMENT APPROACH PROCEDURES

Missed Approach Procedure

- From Missed Approach Point (MAP) to point where a new approach, a holding pattern, or as directed by radar, or entry into en-route flight may be initiated.

INSTRUMENT APPROACH PROCEDURES

The Precision Approach

- A precision approach is one that uses electronic glide path information.
- The Minima for a precision approach is a **DA**.

INSTRUMENT APPROACH PROCEDURES

The Non-Precision Approach

- A non-precision approach is one that does not use electronic glide path information.
- The Minima for a non-precision approach is an **MDA**.

INSTRUMENT APPROACH PROCEDURES

Missed Approach Point

The MAP in a procedure may be:

- intersection of electronic glide path with applicable DA.
- a beacon.
- a fix.
- a specified distance from the FAF.

INSTRUMENT APPROACH PROCEDURES

Circling Approach

The **visual** phase of flight, after completing an instrument approach from which a straight-in landing was not possible, during which the A/C is maneuvered into position for landing, the A/C must not be descended below DA/MDA until visual with airfield and the A/c position can be determined.

INSTRUMENT APPROACH PROCEDURES

Circling Approach

If visual reference lost while circling to land from an instrument approach, the missed approach for that approach procedure must be flown!

INSTRUMENT APPROACH PROCEDURES

objectives:

- Identified the 4 segments of an instrument approach
- Studied the procedures used in the initial approach sector
- Examined the differences between a precision and a non-precision approach
- Studied the circling approach

INSTRUMENT APPROACH PROCEDURES

**ANY
QUESTIONS?**